

ROVANIEMI AP, Finland

WMO: 028450

Lat: 66.562N

Lon: 25.822E

Elev: 637

StdP: 14.36

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-9.6	-19.3	2.0	-14.4	-13.7	2.7	-9.5	25.0	24.1	22.6	23.3	5.7	70	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	76.5	61.4	73.0	59.5	69.7	58.1	64.5	71.7	62.2	69.4	60.1	66.6	9.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	84.2	67.3	59.3	77.1	64.6	57.3	71.7	62.7	29.9	71.5	28.2	69.4	26.8	66.4	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.6	16.5	DB	-18.2	80.8	6.8	3.8	-23.1	83.5	-27.1	85.7	-30.9	87.8	-35.8	90.6	(4)
(5)			WB	-18.3	67.4	6.7	2.7	-23.1	69.3	-27.1	70.8	-30.9	72.3	-35.8	74.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	12.7	13.5	21.7	32.7	43.5	54.2	60.4	55.8	46.3	33.9	23.6	17.4	(6)
(7)		DBStd	18.51	12.04	11.58	8.30	6.52	8.25	7.03	6.05	5.93	6.29	7.60	9.22	11.42	(7)
(8)		HDD50	6308	1158	1022	877	519	237	36	2	14	143	498	791	1011	(8)
(9)		HDD65	11072	1623	1442	1342	969	666	331	168	291	560	963	1241	1476	(9)
(10)		CDD50	748	0	0	0	0	36	160	324	195	33	0	0	0	(10)
(11)		CDD65	38	0	0	0	0	1	6	25	6	0	0	0	0	(11)
(12)		CDH74	220	0	0	0	0	8	43	141	28	0	0	0	0	(12)
(13)		CDH80	23	0	0	0	0	0	6	15	2	0	0	0	0	(13)
(14)	Wind	WSAvg	8.2	8.3	8.6	8.9	8.2	8.6	8.0	7.6	7.4	7.9	8.1	8.2	8.5	(14)
(15)	Precipitation	PrecAvg	21.1	1.5	1.2	1.2	1.1	1.3	2.0	2.4	2.8	2.4	2.1	1.8	1.4	(15)
(16)		PrecMax	28.5	3.7	2.9	2.9	3.3	3.5	4.5	5.2	5.5	4.8	4.3	3.4	2.3	(16)
(17)		PrecMin	15.5	0.4	0.4	0.1	0.2	0.2	0.3	0.5	0.7	0.3	1.0	0.5	0.3	(17)
(18)		PrecStd	3.3	0.7	0.7	0.7	0.7	0.7	1.1	1.3	1.2	1.1	0.9	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	36.8	42.8	55.9	74.5	78.7	82.4	77.6	64.7	51.3	43.0	37.5	(19)
(20)			MCWB	33.8	34.7	36.6	44.8	55.7	61.4	65.4	64.2	56.4	49.0	42.2	36.7	(20)
(21)		2%	DB	32.3	33.2	38.1	50.1	68.2	73.4	77.9	72.8	61.0	48.8	39.5	34.8	(21)
(22)			MCWB	32.0	32.3	33.9	41.3	53.8	57.7	62.3	60.4	54.0	47.3	38.9	34.3	(22)
(23)		5%	DB	30.6	31.4	35.0	46.0	62.0	69.7	74.4	69.1	58.2	46.5	36.9	32.9	(23)
(24)			MCWB	30.2	31.0	31.8	39.0	50.1	56.0	61.1	59.1	53.1	44.9	36.6	32.5	(24)
(25)		10%	DB	28.3	29.0	32.3	42.4	57.2	66.2	71.1	65.7	55.4	44.0	34.5	31.5	(25)
(26)			MCWB	28.0	28.6	30.6	36.5	47.4	54.1	59.7	57.7	52.0	43.0	34.2	31.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	35.0	37.1	46.2	58.0	64.5	68.2	66.4	58.3	50.3	42.4	36.9	(27)
(28)			MCD B	34.5	36.1	41.3	54.5	69.3	73.5	76.1	74.7	62.0	50.7	42.7	37.2	(28)
(29)		2%	WB	32.1	32.4	34.4	42.5	54.7	61.0	65.4	63.3	55.8	47.8	39.1	34.2	(29)
(30)			MCD B	32.3	32.8	37.0	48.1	66.4	69.7	72.9	69.3	59.1	48.7	39.6	34.6	(30)
(31)		5%	WB	30.4	30.8	32.4	39.5	51.3	58.1	63.4	60.9	54.1	45.0	36.5	32.4	(31)
(32)			MCD B	30.6	31.2	34.0	45.0	60.2	65.9	70.8	65.6	57.0	46.0	36.8	32.7	(32)
(33)		10%	WB	28.0	28.6	31.0	37.1	48.3	55.9	61.3	58.8	52.4	42.9	34.2	31.2	(33)
(34)			MCD B	28.3	29.1	32.4	41.4	56.0	63.3	68.5	64.0	54.9	43.9	34.4	31.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	9.8	12.4	12.8	14.0	13.8	13.9	12.9	10.9	7.6	8.0	9.5	(35)
(36)			MCDBR	9.7	8.6	13.5	17.2	21.6	19.0	18.3	17.5	14.6	8.8	7.2	8.3	(36)
(37)			MCWBR	9.8	8.1	10.1	10.6	11.0	8.8	7.8	8.0	8.9	7.4	7.1	8.1	(37)
(38)		5% WB	MCDBR	9.9	8.1	11.0	16.0	19.8	16.0	16.2	14.5	12.1	7.9	7.1	8.1	(38)
(39)			MCWBR	9.9	7.8	8.8	10.3	11.0	8.8	8.4	8.4	8.3	7.3	7.1	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.193	0.261	0.293	0.327	0.338	0.340	0.354	0.337	0.313	0.285	0.346	N/A	(40)	
(41)		taud	1.932	2.110	2.178	2.252	2.456	2.481	2.459	2.488	2.540	2.353	2.677	N/A	(41)	
(42)		Ebn at Noon	66	183	240	263	274	276	267	261	242	180	70	0	(42)	
(43)		Edh at Noon	5	17	26	32	30	30	29	26	19	13	5	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	22	175	628	1163	1499	1642	1539	1119	588	204	41	0	(44)	
(45)		RadStd	2	31	65	101	136	124	147	101	63	40	5	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (40 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page